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RESEARCH

Facing the pain: ethical considerations of AI-based pain detection of farmed animals

Mark Ryan[1] · **Leonie Bossert**[2]

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Abstract

Automated pain detection (APD) is an emerging technology that runs on artificial intelligence (AI) (e.g., machine learning, computer vision, and deep learning) and is aimed at detecting pain in farmed animals. One type of APD analyses the facial expressions of farmed animals, indicated by movements in facial muscles, to identify indicators of pain. This paper evaluates six ethical concerns of APD use in the agri-food sector: (1) harm caused by incorrect diagnosis from APD; (2) harm caused by using drones or robots to collect images; (3) the insufficiency of focusing only on pain for animal welfare; (4) increased alienation within the human-animal relationship due to APDs; (5) the need to focus on tackling already known causes of animal pain; (6) the need to focus on tackling the root cause of farmed animal pain (i.e., animal farming industry). This paper proposes four principles for responsible APD development and use in the agricultural sector: (1) Implement the precautionary principle when there is uncertainty about animal pain; (2) Humans must be kept in the loop on decision-making on animals' health and well-being; (3) Individuals (e.g., farmers) should be held accountable for animal health or welfare problems that are detected by their APD systems but not investigated or treated; (4) An open-source approach must be established to train APDs with high-quality images, ensuring the technologies work as well as possible.

Keywords Artificial intelligence · Animals · Ethics · Agriculture · Pain detection

1 Introduction

Ensuring animal welfare has received growing attention over the past few decades. As there is a growing consensus that, at least, sentient animals matter morally for their own sake, there is also a growing consensus that they should not suffer pain (unnecessarily). While defining animal welfare is a notoriously challenging task, definitions range from (a) basic health and functioning, (b) affective states and (c) the ability of animals to live reasonably 'natural' lives. The dominant view seems to understand welfare as valenced experiential states (e.g., pleasure and suffering). So, at least for approaches (a) and (b), the absence of pain seems to be a crucial criterion for welfare. Pain is an undesirable experience involving sensory and affective aspects, as well as actual or potential tissue damage [95, 110].[1] It is an affective state that responds to 'an unpleasant sensory and emotional experience' [95], p. 1976. The physiology of pain comprises sensory responses to tissue damage, which transmit information from 'peripheral pain receptors (nociceptors)' to the brain via nerve pathways [75], p. 2. Acute pain refers to immediate short-term issues, such as 'injury, infection or inflammation' (ibid.).[2] Chronic pain is more long-lasting and consistent (e.g., osteoarthritis, cancer, and neurological

✉ Mark Ryan
mark.ryan@wur.nl

✉ Leonie Bossert
leonie.nora.bossert@univie.ac.at

1 Wageningen Social and Economic Research, Wageningen University & Research, Wageningen, The Netherlands

2 Philosophy of Media and Technology, University of Vienna, Vienna, Austria

[1] With this, we do not want to claim that every form of pain is necessarily bad in every context. It needs to be differentiated between useless and–in one way or another–useful pain. Several philosophers argue that bad experiences are also important for a good life because without them, one could not really appreciate good experiences (e.g., [86]). For a nuanced perspective on pain cf. [23].

[2] However, some claim that pain is not a unifiable phenomenon [23]. Corns criticises reductionist approaches that reduce pain to be a monocausal issue that always needs to be evaluated negatively.

disease). Pain impacts animals' well-being[3] and quality of life, and can make everyday activities challenging (e.g., standing, feeding, and walking).

However, one of the significant practices that inflict pain and suffering on animals is global animal agriculture [109], as today, most animal "biomass" on Earth is farmed animals [6]. So, if there is considerable consensus that we should improve animal welfare, addressing pain induced in animal agriculture is an important issue. An essential first step to achieve this is identifying when farmed animals are in pain [69], and which practices especially reduce their well-being.

In recent years, pioneering research has emerged to identify pain in farmed animals. These range from cognitive bias shifts from optimism toward pessimism [82], physiological indicators (e.g., blood lactate levels [68], cortisol levels [74, 89], salivary oxytocin [71]), animal vocalisations [24], measuring respiration [94], olfactory indicators [21], measuring heart rate variability [5], gait analysis [19], and facial expressions [76]. This research is also being combined with computer vision, machine learning, and deep learning to automate the process, often referred to as automated pain detection technologies (APD) [15] or automated pain recognition (APR) [20].[4] While there are many different approaches under the rubric of APD, this paper focuses on the use of artificial intelligence (AI) to (presumably) detect pain through the facial expressions of farmed animals, as well as potential ethical concerns that may arise.[5]

While some emphasise the importance of implementing multidisciplinary approaches to APD [83] or implementing ethical practices in APD [20], there has been no explicit ethical analysis of APD to date. It is crucial to evaluate the ethical aspects relevant to APD to ensure that its design and implementation are done in a morally acceptable way to improve current practices [46]. In this paper, we aim to fill this research gap and, from an applied ethics perspective, discuss what responsible development and use of APD should look like. With this, we want to provide initial thoughts that can guide practitioners and policymakers on how to discuss these concerns and how (if) APD can be implemented responsibly. To do so, we build on the widely accepted premise that sentient animals count morally for their own sake, that inflicting pain and suffering on them is morally wrong, and that we have at least some duties regarding domesticated animals to maintain or even improve their well-being.[6]

The first section of this paper provides an overview of current state-of-the-art research in the field. Section two focuses on some of the most significant ethical concerns regarding APD, namely: (1) Harm caused by incorrect diagnosis from APD; (2) Harm caused by using drones or robots to collect images; (3) It is insufficient to focus only on pain for animal welfare; (4) APDs will further alienate the human-animal relationship; (5) The focus should be on tackling already known causes of animal pain; and (6) The focus should be on tackling the root cause of farmed animal pain (i.e., the animal farming industry). Section 3 provides initial thoughts and four principles on what should be considered when designing and using APDs responsibly, grounded in ethical analysis. This paper contributes to the literature by focusing on the (currently underdeveloped) ethical analysis of APD, allowing researchers to incorporate these insights into the development of APD.

2 AI to detect pain in farmed animals

Research has often used human observation to identify and manually code animal behaviour and facial indicators. This research focuses on identifying patterns and changes in animals' behaviour and facial expressions, either directly or through video/still images. This process has many limitations: it requires human training, is highly time-consuming, can be invasive to the animal, and risks human error and bias [16]). Furthermore, these manual checks may not be sufficient because the animal behaves less naturally in the presence of humans.

In response to these limitations, a growing body of research is exploring whether we can use sensors, biometric data, and AI to automate the identification of patterns and changes in animal behaviour and facial expressions from video or image data [84]. The use of AI to detect pain is built upon models that provide insights about pain in farmed animals [16, 20], for example, AI to detect facial expressions demonstrating pain in farmed animals (APD [73]).

It is believed that facial expressions, indicated by movements of groups of facial muscles, reflect internal affective states. These groups of muscles (or 'action units') are representative indicators of pain responses [70]. These facial

[3] For the purpose of this paper, we use the terms animal welfare and animal well-being synonymously. We are, however, aware that the two terms are also used with slightly different meanings. For a more detailed discussion on animal welfare, see [17, 18].

[4] For simplicity, we will refer to both formulations as APD throughout the paper as there does not seem to be significant differences between them.

[5] These applications are primarily developed to be used in industrial animal agriculture and lab animals. They could also be applied to wild animals if trained with appropriate data, and their use for companion animals is an obvious possibility. Facial recognition models for cats and horses already exist. However, in this paper, we will focus on their application to farmed animals. Within most theories of animal ethics, the practice of farming animals is criticized. We address this issue in Sect. 3.6.

[6] We use this as a premise here, as the focus of the paper is not to provide arguments for these assumptions. It is intensively discussed in animal ethics. For many, see [11, 86, 90, 109].

expressions are difficult for the animal to hide and may indicate internal pain in response to stimuli [54]. If these indicators can be identified, the process could be automated through computer vision and AI. The benefit of APD is that pain identification can be done much more consistently, in a more timely manner, and more effectively than in person [54].

Some APDs are based on 'hand-crafted' AI models[7] developed in research, while other models use deep learning (DL) [16, 20]. The hand-crafted approach uses structured labelled data based on previous research on pain-related facial patterns and grimace scales, and machine learning to predict features in facial images [20]. These models are developed to identify specific statistical patterns in the images they analyse. While hand-crafted machine learning can be helpful, it requires human intervention to learn (which could be biased and lead to incorrect results).

The deep learning approach benefits from training artificial neural networks with multiple layers to 'automatically extract hierarchical features from vast datasets like video data. Convolutional Neural Networks (CNNs) are particularly effective for image processing tasks like classification and object recognition, offering superior performance by mapping individual inputs to single outputs' [20], p. 8. In addition, 'animals may have much more significant facial texture and morphology variation than humans. While initially perceived as a challenge, this may be advantageous when employing a DL approach' [20], p. 9. While deep learning offers great potential, it is often less explainable and understandable how the model arrives at its results compared to hand-crafted models.

In addition to the distinction of models, Broomé et al. [16] also identify that research is divided between 'parts-based' and 'holistic' approaches. Parts-based methods focus on dividing 'input data into different areas, e.g., considering different facial features separately, while holistic methods process the information of the input data as a whole, be it at the body or face level' [16], p. 579. Most of the papers that focus on animal farming are parts-based,[8] e.g., focusing on eyes, nose, and ears, or combinations of these parts of the face [16], p. 580.

Building upon this research, Hewitt and Mahmoud [54] developed a machine learning model to handle the profile and frontal faces of sheep, which could detect individual sheep faces and specific facial landmarks that indicate pain [54]. They used a dataset with 850 sheep facial images (the Sheep Facial Landmarks in the Wild (SFLW) dataset) to develop an approach that incorporates landmark localisation to improve automated facial expression analysis in sheep [54]. Their 25 facial landmarks are implemented in their fine-tuned CNN model, which they hope to implement in a 'near real-time video pipeline' [54], p. 22.

A different study [70] developed a taxonomy for sheep facial action units based on the standardised sheep pain facial expression scale (SPFES) [76]. Lu et al. developed an 'automatic multi-level approach for estimating pain level in sheep by extending computer vision techniques widely used in human emotion recognition' [70], p. 394. Their approach demonstrates how pain identification in sheep can be automated using facial action units. The paper demonstrates the usability of their model with many different sheep face datasets. The authors propose that their approach could be extended to other animals in future research [70], p. 394.

Building on the SPFES, also developed a transfer-learning and fine-tuning approach to evaluate sheep facial images. They implement pre-trained convolutional neural network (CNN) architectures to group 'normal' (no pain) and 'abnormal' (pain) sheep images.[9] Their models achieved a success rate of 93.10–100%. This paper demonstrated a method for automating sheep face classification and pain identification (Noor et al. 2020). Overall, there is a growing body of literature and practice of developing and implementing APD to detect and prevent pain in farmed animals.

[7] One crucial research paper on this topic was a study by McLennan et al. [76], which identified facial pain indicators in sheep with mastitis or footrot [76]. Mastitis is a painful inflammation of the mammary glands in animals that, if left untreated, can lead to death. Footrot is a form of tissue damage caused by the bacteria Dichelobacter nodosus, resulting in pain and lameness in the animal. McLennan et al. [76] photographed 73 sheep with footrot and 17 with mastitis (alongside control groups) to determine facial indicators in these animals and changes over several weeks with treatment (e.g., antibiotics or anti-inflammatories). The researchers detected evident facial movement changes associated with the painful diseases of footrot and mastitis: orbital tightening, cheek tightening, abnormal ear positioning to the front or the side, abnormal lip and jaw profile, and abnormal nostril and philtrum shape. These indicators ranged from 'not present', 'partially present', and 'present'.

[8] While there are several papers on holistic approaches, most are not from animal farming (e.g., horses and cats, instead). While parts-based approaches risk being too narrow in their account (ignoring the close and important connections between body parts in regard to health and well-being), we do not discuss this here, as it requires a paper on its own.

[9] For some individuals, being in pain is their 'normal' state, and we do not want to perpetuate the perspective that these individuals are 'abnormal'. Here, we only describe Noor et al.'s approach, using quotes to highlight the controversial normativities underlying the terms normal and abnormal in this context.

3 Ethical considerations of using AI to detect pain in farmed animals

This section applies an applied ethics approach to address pressing ethical concerns related to APD.[10] One central task of applied ethics is to be action-guiding and provide reasons for how (not) to act in real-world scenarios. As real-world scenarios are embedded in pluralist societies with people building on various values and world views, we do not want to provide arguments on how a specific line of ethical thought (e.g., utilitarianism, deontology, or care ethics) would evaluate APD technologies, but rather provide reasons that can serve as a normative orientation for every practitioner or developer, regardless of what ethical school of thougth they (in some cases intuitively) stick to. In this section, we highlight the ethical challenges posed by these aspects and offer initial thoughts on what this should or could mean for the responsible development and deployment of APD technologies.[11]

3.1 Harm caused by incorrect diagnosis from APD

One of the most fundamental requirements for using APD is that the results must be as accurate, reliable, and implementable as possible. Using APD should allow the identification of pain in a timely, effective, and transparent way. However, many technical challenges are involved in providing accurate, reliable, and implementable recommendations, which also pose ethical concerns. If APD's recommendations are based on shaky foundations, animals could be underdiagnosed, overdiagnosed, or misdiagnosed.

If the APD gives incorrect results, it could lead to *underdiagnosis*, which would result in animals suffering in pain as they are left untreated. Undetected diseases could spread and harm other animals on the farm, neighbouring farms, or in the wild. Overall, pain levels could increase for individual animals and the herd, which—obviously—needs to be avoided if animal welfare should be maintained or, at best, improved. Underdiagnosing due to the false application of APD risks an (unwantingly) disregard of animals' interests in not suffering, thereby failing to meet the goal of sustaining or even improving their welfare.

If the APD *overdiagnoses*, then there is a possibility that medicine and antibiotics will be prescribed incorrectly. This overdiagnosis may have various ethically challenging 'side-effects', e.g., it could lead to antimicrobial resistance (AMR), complicating infections in animals and posing a risk to human and nonhuman health through transmission from food, direct contact, or contaminated environments. Overdiagnosing due to the false application of APD can have negative consequences for individual animals' health, human health, and ecosystem functioning, all of which should be avoided for good reasons (for a detailed discussion, see [88]).

If APD *misdiagnoses*, it incorrectly determines an animal's condition or does not identify it at all. This misdiagnosis of the animal's condition could delay treatment, lead to the administration of incorrect or harmful medicine or therapy, and exacerbate the animal's pain [22]. It could also lead to unnecessary actions at a herd level, extending incorrect procedures and treatment, and subsequently increasing pain for the wider herd, i.e., for more individual animals. Again, if farmed animals' interests are taken seriously, such misdiagnoses due to the false application of APD must be avoided, as they risk setting back their interests.

So, APD technologies must be based on sound evidence and methods to reduce animal pain and suffering. In the remainder of this section, we discuss three practical and theoretical challenges to the veracity of APD: image quality and quantity (1), ground truth (2), and interpretation bias (3).

Firstly, collecting images of animals is difficult because of the visibility of the animals in sheds, issues involving the camera (e.g., dirt or rain on the lens), lighting or darkness in the area, and the difficulty of obtaining images in open spaces (e.g., there is a limited number of places to fit cameras, and the animals may be too far away from the camera) [20] (1). Animals are often moving, and capturing clear facial images can be technically challenging, especially when background movement is excessive, which is typical on farms. Animals also block cameras, move, and damage them [87].

Therefore, obtaining sufficiently high-quality facial images of farmed animals significantly undermines APD's effectiveness. A great deal of time, money, and effort goes into obtaining images of animals on farms to train APD algorithms. However, if these images are still of poor quality, it may lead to poor recommendations. Effective APD also depends on the availability of large, labelled datasets for training its models [54]. In human facial recognition, AI systems are trained with vast amounts of facial images from different angles, resolutions, backgrounds, compositions,

[10] While there are many deep questions and assumptions related to APD and their application (for example, whether facial expressions actually reveal pain, whether translating facial expressions is an over-simplification of animal emotion, and risks of anthropomorphism and epistemic bias), these are beyond the scope of this paper. The purpose of this paper was to provide an applied ethics evaluation of APD, which is currently missing from the literature. This evaluation can be helpful to practitioners in the field developing these technologies and to policymakers implementing guidance and regulation for APD.

[11] While the ethical challenges we discuss here are interlinked and sometimes difficult to separate, we nevertheless present them under separate headings to improve readability.

and people. For example, human facial datasets such as MegaFace include '4,753,320 faces of 672,057 identities from 3,311,471 photos downloaded from 48,383 Flickr users' photo albums' [51]. This large repository of images is not the same for farmed animals. For example, one of the studies analysed used only 600 sheep facial images (with eight landmarks) to train their APD models [120]. An open practical question is how to collect these high-quality animal images without disturbing them (e.g., could photographing them serve as a form of enrichment and counter the boredom many animals face in animal agriculture? See also Sect. 3.2). On a more theoretical level, if taking animals' interests seriously, the question also needs to be addressed whether animals could have an interest in privacy [92] and whether methods must be found to get an animal's consent before photographing them (e.g., only taking photos if the animal voluntarily participates in this practice).

Even if the images are high quality, a large number of diverse images is needed to ensure the models do not lead to overfitting (e.g., the training data sample is limited and does not have enough different representative samples for a range of input data values—age, gender, and breed of the animal) [15]. There may be variations in facial pain indicators across these factors or even among individual animals. For a model to be effective at identifying facial indicators of pain, it must be tested across robust datasets to avoid biased results and recommendations [27]. As in the human case, animals could also be affected by gender or age-related biases inherent in the technology. This obviously conflicts with addressing individual animals' interests, rather than treating them as a herd or population. While the social embeddedness of animals is morally relevant, an animal-centred ethics perspective focuses on the individual animal and requires responsible APD technologies to foster individual animals' well-being.

A second concern about APD is the lack of ground truth (2). It is easier to identify pain in humans because we can ask the person; however, assessing pain in humans also comes with various challenges [99]. This self-reporting is not possible with animals.[12] It has been shown that facial expressions (in humans) convey certain emotional states, such as pain, so there is an assumption that this also occurs in non-human animals, at the very least, other mammals [33]. Therefore, it is assumed that at least some animals express (some forms of) pain through facial expressions [41]. However, if this assumption is valid, it may also be the case that animals' facial expressions relate to other affective states (other than 'pain'). Other affective states, such as fear and stress, may also be evident in animals' facial expressions. The challenge of differentiating and classifying pain, as opposed to other negative affective states, should be made transparent with a caveat about the limitations of the ground truth. One way to do this could be to classify and group facial expressions elicited by stressful, painful, and harmful stimuli as 'negative indicators'. Even if the specific affective state being exhibited is uncertain, there is a clear indication that it is not a positive or an at-ease state. This information about animal states would still be valuable in responding to their unease. While it is undoubtedly challenging to pinpoint the exact *type* of affective state of an animal, grouping them into broader categories, such as negative and neutral/positive facial expression indicators, may help overcome some of the challenges of inferring specific affective states.

Thirdly, identifying animal pain is often reliant on human interpretation [20], p. 9 (3). For example, animal grimace scales rely on human researchers to manually annotate their interpretation and scoring of animal grimaces. However, this is prone to 'the risk of bias and subjective judgment' [20], p. 3. In other cases, relevant data may be omitted from APD research because they are challenging to include. For example, Mahmoud et al. [73] developed a sheep facial AU taxonomy based on the standardised sheep pain facial expression scale (SPFES, [76]), but used only frontal facial features, such as ears, eyes, and nose. They omitted cheek and lip profiles (initially in the SPFES) because those features 'can hardly be seen on a frontal face' [73], p. 9. They also only defined two pain levels (pain or no pain), differing from the original distinction found in McLennan et al.'s SPFES—namely, pain being 'not present', 'partially present', and 'present' [76]. When APD research omits specific pain indicators because they are too tricky, it runs the risk of not accurately identifying pain, thereby risking being applied incorrectly, contrary to maintaining or improving animal welfare.

Essentially, if APD models are not trained on high-quality and robust data, they are likely to lead to errors. As a result of these errors, farmers, veterinarians, and agri-food workers may receive incorrect information and advice. Therefore, APD developers must use a large number of high-quality images to train their algorithms. Attention must be paid to retrieving robust datasets and testing APD models to ensure accurate representation. It is essential that APDs are not seen as providing unequivocal pain indicators, and that other (non-digitalised) actions be implemented to supplement or replace APDs when apparent shortcomings in the data are apparent.

However, classifying facial expressions and measuring the pain of another living being are, in themselves, highly interpretative exercises, whether automated (e.g., APD) or non-automated (e.g., observations using annotated grimace

[12] While some animal ethicists claim that communication with animals would be possible if we allow more creative forms of communication [37], we are still far from establishing such forms of communication.

scales). This challenge is evident in other humans and is even more difficult with animals because they cannot communicate their experiences to us in the same way other humans can [41], (but see [77] for more on animal communication). When identifying the states of well-being of another living being, we rely on interpretations [81]. Therefore, there is a tension between using APD to reduce pain and the fact that the pain of another living being can never be fully known. As this tension cannot be eliminated, APD researchers should acknowledge this limitation in their research, and the results from APDs are based on current scientific knowledge on facial expressions and affective states. So, no matter how good the technology is, there is always a degree of *uncertainty*.

3.2 Harms caused by retrieving images to train APD

As discussed above, retrieving animal images for training APD algorithms can harm or disturb animals. Here, we want to focus on two specific cases: the physical interference of animals on farms through drones and robots retrieving images, and the need for more images, which necessitates additional animal testing.

3.2.1 Harm caused by robots and drones obtaining images for APD

Using robots and drones to obtain images (for use in APD) may be a more effective way than stationary cameras because the latter are often 'difficult and expensive to implement in exceptionally large farms' [84], p. 15. Robots and drones make it easier to retrieve images because of their mobility. They can perform the function of several stationary cameras due to the distance, movement, and large number of animals involved. This option may also be cheaper because farmers invest in only one drone or robot rather than numerous cameras [84].

However, they can pose a risk to the lives and well-being of farmed animals by disturbing them, causing stress, and even posing physical danger [101]. On a theoretical level, there is the possibility that animals trip over, become entangled in, or are struck by drones and robots, causing them discomfort, pain, and injury [100]. These devices may also startle or scare the animals, leading to elevated heart rates, stress, anxiety, and fear. The animals may also react by bumping into each other, falling, running into objects, and hurting themselves [102]. These risks must be taken seriously if animal interests are to be taken seriously, and the impact of autonomous devices (as a means of retrieving images for APD) must be explored [10]. Therefore, it is essential to evaluate current scientific research to identify the impacts of robots and drones on farmed animals, and we will briefly highlight some of these findings in what follows.

While technical literature is abundant discussing various deployments of robots and drones in animal farming [2, 3, 7], there is minimal investigation into the emotional and physical harm they may cause. Some studies have evaluated the impact of robots on chickens. For example, one study used ground robots to collect floor eggs in cage-free housing systems to prevent contamination or breakage [67]. The study also measured 'the effects of ground robot manipulation on production performance, stress response, bone quality, and behavior', and concluded that the robots did not negatively affect hen wellbeing [67], p. 1).[13] In the context of cows, a study examined the physiological and behavioural response of dairy cattle to the introduction of robot manure scrapers [34], concluding that there were signs of minor disturbance when the robots were introduced in the first week, but this dissipated once they became accustomed to their presence [34]. Another study analysed cows' behaviour in response to pasture sanitation robots, noting that they appeared more comfortable interacting with stationary robots but fled from moving robots. Some literature points to improvements in animal welfare with greater automation and a reduction of human contact [65].[14] This suggests that ethical evaluations must vary depending on which species or individuals are affected and whether they benefit from respectful contact with humans—for example, because it provides them with variety and entertainment, or because it causes them anxiety.

Since robots and drones often interact closely with farmed animals, they must be developed to ensure they do not cause harm during implementation. Suppose there are clear indications that using robots and drones (for retrieving images for APD) is highly harmful to the health and welfare of specific farmed animals. In that case, these methods should be reconsidered in favour of less harmful alternatives, such as stationary cameras. However, further research is needed on how robots and drones acquire images for APD, the potential harms to different animal species, and whether they can be designed to mitigate these harms (for a review of animal-robot interactions, cf. [98]).

Furthermore, additional research is needed to evaluate the types and levels of invasiveness affecting animal health and welfare. This research is essential to determine whether a certain level of mild aversive experiences is deemed

[13] Similarly, other studies found that ground robots did not induce greater stress than humans in broiler houses [32, 91].

[14] For example, one study implemented drones to herd sheep and concluded that 'Sheep acclimatised quickly and positively to the drone initiating drive of a flock, regardless of drone speed' [121], p. 1. Another study that focused on drones for herding cows demonstrated that the cows were unbothered (and ignored) the drone after the first day of implementation [4], allowing the interpretation that the drone does not disturb them at all.

acceptable to yield important health and welfare outcomes for those individual animals (by identifying and preventing pain, ill-health, and disease), and for other animals (presumably, by training APDs that can be used on similar animals around the world), and arguments for or against this must be provided.[15] However, as long as no fully fleshed arguments in the context of APD exist—arguments that ideally should be widely agreed on by animal advocates, policy makers, and other stakeholders, researchers must err on the side of caution and use less invasive methods to obtain sufficiently high-quality images to ensure the technical robustness of APD. This is ethically mandatory to avoid potentially unnecessary harm to animals.

3.2.2 Additional animal testing for pain metrics and images to train APD

The need to collect sufficient animal images may require conducting further research to identify pain metrics and obtain more images of animals in pain for training APD algorithms. Most APD research is based on previous (non-automated) research that identified and categorised pain expressions. This research is typically conducted by observing and manually annotating the animals' facial pain. One of the foundational methods to do this was the 'Mouse Grimace Scale' (MGS) [61–64]. The MGS was first developed to determine the pain felt by mice through changes in their facial expressions, such as orbital tightening, ear changes, and whisker alterations [20], p. 2. Often, pain-inducing stimuli is applied to the mice (and other animals), a procedure that conflicts with the principle of not harming or inflicting pain on animals.[16]

In addition to these earlier approaches to measure grimace scales, scientists often develop pain diagnostics with three experimental groups: untreated controls, animals with postsurgical or other untreated pain, and animals whose pain is being treated [41, 42]. Sometimes, a fourth group is included: animals that do not undergo painful procedures but receive analgesics to determine if the analgesics themselves make the animals feel ill. To improve understanding of animal pain, how it is exhibited in animal facial expressions, and to better train APD algorithms, researchers may conduct more experiments in which they inflict pain on animals to observe their responses, or create studies in which certain animals are left untreated [41]. This painful stimulus has come from, e.g., castration [25], tourniquet application and capsaicin injection [44], osteotomy [50], ear tattooing [57], tail-docking [49], ear-tagging and microchipping [72].

Therefore, the concern is that scientists will implement more of these painful procedures to develop pain metrics for APD models and/or obtain images of animal pain for training these models, thereby setting animals' interests back and reducing their welfare.[17] There is also the possibility that pain medication will be unnecessarily withheld from animals to evaluate their pain reactions. This is ethically problematic because it allows the continuation of suffering for the sake of scientific research, where there are easy remedies to alleviate it [41]. In addition, some researchers may even intentionally cause suffering to retrieve images and to train the AI to detect pain [62]. These are serious and problematic risks for the development and adoption of APDs in practice.

Such practices appear to be prima facie in contradiction with the purpose of APDs—to reduce pain in farmed animals. However, this depends upon the ethical framework that one adopts. For example, suppose one approaches it from a deontological standpoint. In that case, one may argue that these outcomes are fundamentally wrong and that APDs should be developed and implemented only when they reduce pain, or at least do not increase it. Even deontological approaches that allow rights violations in extraordinary cases to prevent the violation of a greater number of rights (e.g., Abbate 2020) would most likely not permit such practices, as they do not count as extraordinary cases of conflict. However, a utilitarian may argue that the benefits of developing technologies that can be used across farms worldwide to reduce animal pain outweigh the harms of withholding medicine to animals in pain, implementing painful procedures to obtain images for AI training, and purposefully causing pain to a few animals.

While deontologists and utilitarianists might hold different views on whether inflicting such pain could, in some cases, and under very specific circumstances, be morally permissible, we think that the use of APDs on farmed animals can only be classified as 'responsible' when it is used in a way that takes animals' interests seriously, and that sustains or even increases their well-being. While on an abstract theoretical level, it might be possible that conducting such research on a few animals could lead to results that will overall benefit more animals, on a more concrete and practical level, this means continuing practices that harm animals to produce technologies that are used in a context that is somewhat hostile to animals (animal agriculture). Even if these technologies could be a means to make it a

[15] A consequentialist account would differ in its evaluation of this question from a deontological account, and virtue ethics or care ethics may come to even different results. As we do not want to argue for a particular ethical view in this paper, we aim only to highlight that arguments for prioritisation and for balancing benefits and harms at different levels must be provided in future research.

[16] Since the development of the MGS, grimace scales have been developed for 11 species of animals (e.g., rabbits, pigs, cows, rats, and sheep) [20].

[17] Thanks to an anonymous reviewer for pushing us to address this issue.

little less hostile, inflicting pain on animals to achieve this aim will be very hard to justify ethically. So, although this requires more research, we hold that, even from a utilitarian standpoint, inflicting pain on animals, e.g., to obtain more high-quality images of animals in pain to train APDs with, should be seen as morally impermissible, as it is from a deontological and a care ethics perspective. Therefore, developers and researchers must refrain from doing so.

3.3 Focusing only on pain is insufficient for animal welfare

Another concern about APD, which is inherent to the technology and cannot be technically solved, is that it focuses only on the unfavourable conditions of animals, namely, pain detection. While pain is an essential consideration for farmed animal welfare, concentrating on it may overlook positive states and emotions [13] that are just as crucial for animal welfare [8, 9].

Within animal ethics, there is a broad consensus that animal well-being is a rich concept. A normatively sophisticated concept of animal well-being, which is linked to an animal being able to live a good life (understood colloquially, not as an eudaimonistic philosophical concept), recognises that an individual's well-being requires more than the mere fulfilment of basic needs. Basic needs encompass adequate food and water, protection from heat or cold, and freedom from pain, suffering, or stress, and are relatively easy to measure. However, a normatively sophisticated account of a good animal life must go beyond these quantifiable aspects [86]. There are at least two reasons why this broader perspective is essential [13].

First, disregarding the range of positive states and emotions that animals can experience perpetuates a reductionist view of them, as if the absence of pain or suffering were the only thing that mattered. We would reject such a simplistic conception of a good life for humans, acknowledging that aspects such as political and social participation and education are integral to it. Similarly, some animal populations exhibit cultural practices [111], and participation in these practices may contribute to their well-being, even if such contributions are difficult to measure.

Second, this issue also concerns the societal relationship between humans and animals and therefore extends beyond the perspective of individual animal well-being. After decades of animal-centred ethics approaches highlighting the moral shortcomings of prevailing human treatment of sentient animals, there is a need for positive visions of what human–animal relations ought to look like. This topic has been the focus of intense debate in recent animal ethics [35–37, 104]. Developing an understanding of what constitutes a good life for other animals is essential for shaping such visions—and, again, should go beyond the need to be free from pain to do justice to sentient animals.

To support this ethical standpoint, there have been extensive scientific studies into various species of animals to demonstrate that they have complex emotional repertoires, they fear, love, hold grudges, create communities, and engage in cultural practices [28–30, 111, 113]. In the context of farmed animals, many studies have also demonstrated variations in group dynamics and well-being indicators beyond pain. For example, sheep (the most commonly studied animal in APD) form stable social groups, and isolation from these groups causes them distress [40], to graze less [105], have weaker immune recovery [31], and poor adaptability afterwards [85]. Other studies have demonstrated the importance of play for farmed animals in their groups. For example, cows [78], sheep [52], and goats [59] exhibit play behaviour such as mock fleeing, mock aggression, environmental exploration and mock copulation [93].

This acknowledgement of scientific research done on the different components required for animal well-being has been taken up at an international level with the establishment of the EU Platform on Animal Welfare in 2024. The subgroup on animal welfare policy indicators aims to help the European Commission to 'identify meaningful indicators in line with EU animal welfare policy objectives and the corresponding methodology to collect, consolidate and interpret them for policy purposes' (European [39]). The allocation of freedom from pain is only (a part of) one of the five indicators [115].[18] While concentrating on pain is essential, it should be one factor in an overall holistic approach to animal welfare rather than the sole condition. Given that an EU policy group has already brought this forward, we consider it an important entry point for a growing awareness of the complexity of animal welfare, one that can and should be taken up by stakeholder groups in the context of APD development. Since this problem is inherent to APD technologies and cannot be solved through higher image quality or a more diverse image selection alone, we consider it crucial to take it into account during both the development and, in particular, the application of these technologies. One initial practical way to address it in application could be through the use of a disclaimer. Just as AI chatbot companies point out that AI can make errors, an APD system could indicate that factors beyond pain—ones the system cannot detect—should also be considered to obtain a more comprehensive

[18] The indicators discussed in this subgroup are freedom from hunger and thirst, freedom from discomfort, freedom to express normal behaviour, freedom from fear and distress, and freedom from pain, injury, and disease [115]. While all of them are highly important, the focus on 'being free from something negative' still seems to be reductionist, as a philosophically and biologically sophisticated understanding of animal well-being also requires experiencing positive states, like joy, love, companionship, and not only being free from negative states.

understanding of the animal's well-being. Also, like pain, many positive states can be measured via biosensors (e.g., by the amount of the happiness hormone serotonin in the blood). This research could complement APD, and we intend to spark discussion on whether complementing APD with technologies that measure positive states would be an ethically promising approach.

3.4 Further distancing of the human-animal relationship

When calling for greater attention to animals' interests, e.g., their interest in avoiding pain, APD could be used to implement this by enabling earlier pain detection. There is potential for APDs to provide greater care and attention to animals while helping farmers address their health and welfare. However, APDs may also contribute to further automation in the industry, shifting more tasks within the agricultural system to technology. Although the concrete societal effects of a large-scale implementation of these systems have yet to be thoroughly researched, some assume that delegating another care-related task (recognising if an individual is in pain) from human hands to AI-based technologies is accompanied by an ever-increasing automation of the animal agricultural industry, which may harm human-animal relationships [14].

There is a risk that removing human expertise will harm animal welfare [12]. While efforts to use APD seem well-intentioned (e.g., to improve animal welfare), they may still perpetuate a system of alienation between humans and animals. Rather than mitigating this disconnect, technological interventions in agriculture can be seen as deepening the estrangement of animals used for farming. However, alienation from farmed animals is not solely a product of AI-driven technologies (such as APD), but it has long been a feature of industrialised animal agriculture. However, AI systems (of which APDs are one) may further disrupt these interactions by replacing the human caregiver with machines or technological systems [14].

Some propose that 'AI threatens to make genuine care even more difficult to achieve' in animal agriculture [107] as it eliminates what (often little) direct human-animal interaction remains. In less automated farming systems, the care relationships between farmers and animals in intensive agriculture may be questionable (depending on one's understanding of 'care'), but this bond could be further eroded if machines take over even more tasks [116]. This shift will likely affect farmers and extend to their families, farm workers, veterinarians, and others involved in animal agriculture, influencing human-animal relationships on a broader societal scale.

While we think this is a significant criticism of technology's impact on (humans' perspectives of) animals, and provides many reasons to be cautious, we also want to stress that this is not necessarily specific to APD, and, as with the ethical challenge we discussed in Sect. 3.3, it is not easily addressed by APD researchers. Instead, this topic needs to be discussed on a broader societal level. Ideally, it should be discussed whether societies want to continue shifting towards even more automated animal agriculture (which they are already moving towards) or if they consider this a problem. We consider it essential that these discussions accompany APD development and deployment before it is broadly established. Currently, society can still oppose a large-scale implementation of such technologies. Reducing the use of established technologies is often much more difficult than not introducing new technologies on a large scale.

Furthermore, one may object that it is still largely unclear whether the introduction of APDs will, in the first place, lead to the further alienation of animals, the distancing of the animal-human relationship, and the (further) reduction of animals to mere objects for our consumption. While this argument is usually levelled at AI and the further automation of the animal farming industry, it is unclear how APDs themselves would specifically cause this. This is because it appears to contradict one of the central aims of APDs: to identify more incidences of animal pain and stress, rather than fewer, and thus, respond to them. APDs could do this because it is currently challenging to identify and respond to pain due to farmers' overburdened workloads, the large numbers of animals, and the difficulty of detecting pain in animals.

One may also object that there are currently no indications that APDs will completely automate interactions between humans and animals, as they merely enable the identification of illness or suffering while still requiring the farmer to intervene and take appropriate measures. Therefore, the concern that the APD technologies themselves will further distance the human-animal relationship appears somewhat at odds with current and likely (near-future) scenarios. However, suppose the entire process is automated, with the farmer removed from prescribing and administering medicine and care. In that case, this risk clearly exists and would be problematic for the animal-human relationship. This certainly needs to be addressed in guidelines for the responsible implementation of APDs.

3.5 Focus on current known causes of animal pain instead

Another criticism against using AI to detect pain in farmed animals is that time and resources could be better spent tackling (already known) causes of animal pain. Even though AI may help reduce pain in farmed animals, there are already so many situations where animals are in pain that can be tackled now. This argument emphasises an ethical obligation to address tangible, current pain-related issues in farmed animals, rather than hoping that APD will eventually reduce farmed animal pain. There is no need for an advanced machine learning algorithm to realise (and respond to) the pain of animals. Disease, poor housing, painful routine practices, bad handling, and parturition cause pain.

Firstly, *disease* can originate from many sources, such as infections, injury, hereditary defects, the animal's environment, or the intensification of production processes inherent to animal farming. While the causes of disease vary, the indicators of pain resulting from disease are often quite similar. For example, lameness could be caused by infectious agents, injury, poor environmental conditions, or heredity. However, the signs of lameness are often similar (difficulty standing and walking, and significant time spent lying down), so identifying them would allow farmers to respond to these issues.

Secondly, certain types of *animal housing* can be painful to walk on or cause slippage, resulting in injury and subsequent acute or chronic pain. For example, slated flooring can cause bursitis and skin lesions and increase the chance of lameness [53]. Animal housing is often designed with right-angle bends, which make it difficult for animals to walk and cause falls, bruising, and pain [118]. Floors are often poorly maintained, which causes trips, collisions, and falls (ibid.). Fences and gates may also be poorly designed, leading to animals becoming stuck, cut, or grazed.

Thirdly, *routine practices* carried out on farms often cause pain. These can include castration, tail docking, dehorning/disbudding, ear tagging/notching, branding/freeze branding, mulesing, teeth grinding/clipping, nose ringing, and beak trimming [109]. Poor handling of animals can also cause them pain,for example, tail-twisting, pushing, hitting, kicking, prodding, and other physical methods used to move animals often cause pain [26]. These methods can cause animals stress and anxiety, which may lead them to injure themselves on walls, fences, or stables, and to slip or fall [106].

Fourthly, *giving birth* causes pain for farmed animals, but there is a lack of pain relief given to them [75], p. 6. Many farmed animals may endure even more unnecessary pain for extended periods because of a lack of pain medication given to them.

One might object that addressing these sources of pain is a better use of time, resources, and effort than developing and implementing APD. For example, disease is a significant cause of pain in farmed animals, so it is essential to identify when animals have a disease, determine the specific disease, understand the level of pain it will cause, and treat it (if possible). It is also essential to understand the different diseases animals can develop and to implement precautionary measures to prevent them from contracting these diseases in the first place. However, APD is being developed to support current efforts to identify and respond to disease in animals rather than replace them, and this complementary use must be secured for an APD to be responsible. Regarding the poor conditions and housing of animals, a (relatively) easy remedy for the flooring issue is installing better flooring that, at least within the system, better meets animals' interests. Overall, adequate maintenance of flooring, fences, gates, and the environment for farmed animals can be ensured, and the environment can be designed to reduce pain. While many of the handling issues that cause pain can be overcome with 'simple changes in the design of facilities, or improved education of handlers' [75], p. 3[19] and pain caused by parturition can be relieved through medication.

However, while these responses are essential and should be implemented to reduce farmed animal pain, they are not mutually exclusive with using APD. Efforts to reduce many of these incidents that set back animals' interests can also be realised alongside APD research and implementation. In fact, if developed and deployed correctly and taken seriously, the use of APDs may identify and draw attention to these causes of pain (by identifying signs of pain), thus allowing them to be more easily identified and addressed. In addition, most APD research is conducted in smaller projects, universities, and research institutes. These do not directly compete with or divert funds that would otherwise be spent on preventing painful animal procedures or the mistreatment of animals.[20] Many of the causes of pain listed in this section can be reduced through better legislation, improved education, and often, more compassion for the farmed animals themselves. These approaches are not typically in competition with APD for resources, so we do not see a reason to assume that one must choose between them or APD to reduce animal pain. It is not a zero-sum trade-off

[19] Of course, they could also be overcome by ending factory farming, as has been called for in important works in animal-centered ethics approaches for decades [60, 97, 108]. We touch upon this issue in Sect. 3.6. In addition, not only animal-centered ethics approaches call for a reduction of industrial animal farming, but also sustainability and climate ethics [58, 119]. A shift to more plant-based food systems and agriculture is argued to be able to address current global crises such as climate change and biodiversity loss [112].

[20] If it were the case that large amounts of funding were being divested away from other, maybe more effective, ways to reduce pain toward AI pain detection, this could be a good reason to call for a moratorium on researching and developing APD technologies. But that is not the case. Research is usually carried out through research institutes and the private sector, and in our non-ideal world, it seems highly unlikely that they will spend the same money on animal charity or research on alternative methods that are not as profitable as AI systems.

between allocating investment to other pain reduction methods and the development and use of APD.

However, for APD to be applied responsibly, it must be used to address the issues just mentioned rather than perpetuating them. This suggests that APD should be used as a supplementary tool and not as a techno-fix that only addresses the symptoms of pain rather than the root causes [103]. APD should also not be seen or used as a replacement for disease prevention and veterinary practices. From an ethical standpoint, APD can be used to detect painful diseases more quickly, helping reduce animal pain in cases of poor housing, handling, harmful practices, and inadequate pain medication during parturition.

3.6 Tackle the root cause of farmed animal pain

As we have just seen, many of the root causes of pain in farmed animals stem from within the system and are amplified by the intensification and industrialisation of the animal agriculture sector, as well as by the conditions animals endure. Billions of farmed animals are bred and confined in overcrowded, unsanitary conditions. They are slaughtered after living only a fraction of their natural lifespan, often without prior stunning [47, 48]. This process drives industries producing meat, dairy, eggs, fur, leather, wool, and down, despite the pain and suffering inflicted on the animals [79, 109]. In addition, the genetic modification of farmed animals is one of the leading causes of animal pain. For example, broiler chickens have been so intensively modified that they often cannot support their weight, developing skeletal pathologies, leg weakness, and lameness [43, 55, 117]. These birds sit more often to alleviate pain, which increases the possibility of developing dermatitis lesions, which are also very painful [38, 45].

Another issue is deeply rooted in the historical changes and industrialisation of animal farming over the past several decades. Smaller farms have been declining, often being swallowed up by large farms or agribusinesses, resulting in fewer farms with much larger numbers of animals per farm. As a result, 'these massive farms have heightened the challenges of identifying, monitoring, and caring for large groups of animals. This scale has also made keeping the animals satisfied mentally (…) more difficult' [84], p. 1.

Therefore, we think the risk must be taken seriously that the development of APDs may align more closely with the interests of the animal farming industry than with the welfare of affected animals, thereby subordinating animals' vital interests to humans' economic interests. So, an objection to the possibility of establishing a truly responsible use of APDs might be that intensive farming is too profitable and powerful to change radically. The continuation and business-as-usual of intensive animal farming is problematic from both an animal ethics and a sustainability and climate ethics perspective (cf. footnote 18) [56, 58].[21] Therefore, one may argue that efforts should focus on tackling or altogether abolishing the animal farming sector rather than spending time on APD, which does not lead to a radical overhaul of an exploitative and harmful industry.

However, abolishing the animal farming sector or developing APD to reduce animal pain in the industry are not mutually exclusive. There is often more than one sound and convincing solution to ethical problems [80]. There is room to use APDs responsibly to prevent harm and improve animal well-being while simultaneously arguing that the broader practices of animal farming are problematic.

One may support abolishing the animal farming industry, but be aware that this is not an immediate possibility. While it may be a desirable future scenario, efforts must be made here and now to reduce harmful practices, suffering, and pain. Therefore, one can aim for this future vision while also addressing current challenges and issues within the sector. Changing societal practices takes time and requires various activities, efforts, and positions that demonstrate the problematic status quo and explore ways to build bridges between future abolitionist scenarios and the steps to achieve them [66].[22] If applied responsibly, APD can alleviate unnecessary suffering while being one step along a continuum of change in the sector's perspectives and actions. As animal agriculture currently inflicts manifold pain and suffering on farmed animals, we think that reducing that suffering should be welcomed from an animal-centred perspective, even if it is applied in the current non-ideal situation.

4 Conclusion

In this paper, we discuss various ethical challenges associated with using APDs to detect pain in farmed animals. We highlighted the harm caused by image collection, inaccurate results, and challenges related to image quality and quantity. In addition, we addressed the risks of APDs' concentration on pain metrics and the possibility that APDs may further automate the sector, leading to alienation in

[21] As we are focusing on farmed animals, we do not discuss in detail how to evaluate APD technologies from an ecological or sustainable AI perspective, but see van Wynsberghe [114] for that.

[22] In political philosophy, an important differentiation regarding this issue is John Rawls [96] differentiation between ideal and non-ideal theory. The non-ideal theory addresses the imperfections of the real world that philosophical approaches should be able to connect to. While for many animal ethicists in an ideal world, we would not use animals as resources anymore, the non-ideal status quo is one of huge animal farms for which—in the context of a non-ideal theory—arguments can be made as how to use technologies to enable a better life for animals within this non-ideal system.

the human-animal relationship. This paper also explored criticisms that more focus should be placed on alleviating known examples of animal pain or on abolishing the animal farming sector as a whole. In this section, we conclude by outlining the steps needed to ensure responsible use of APD and proposing a set of principles to guide its design and use.

Several aspects must be addressed to ensure that APDs are developed and deployed responsibly. A large and diverse dataset is needed for practical AI training, but such datasets are often lacking for farmed animals. Therefore, researchers must be aware of and transparent about the limitations of their studies when data are limited, and they must provide indications of the uncertainty of their findings. Similarly, researchers should be cognizant of issues such as missing 'ground truth' and the possibility of interpretation bias, and clearly document these in their findings. And while the literature indicates that researchers are building on one another's datasets and research, greater effort could be made to pool data collection and create open data repositories. More pictures of farmed animals could be taken, or the addition of cameras for real-time image retrieval could help obtain better-quality, more robust data.

Regarding the deployment of image-capture technologies, care must be taken to ensure that robots and drones do not cause physical harm to animals, e.g., by crashing into them or impeding them. These devices need to be designed to minimise harm to animals. While the safety of these technologies is still being tested, stationary cameras could be deployed in the meantime.

Additionally, APD researchers should be clear that pain is not the only, or even the most significant, indicator of animal welfare. It may be worthwhile working alongside other researchers on animal welfare to gather more holistic perspectives on the well-being of farmed animals through collaborative research, engagement, and policy recommendations. APD research should not work in a silo of animal welfare.

While APDs could foster attention towards farmed animals as a result of faster identification of pain, illness, and disease, the concern must be taken seriously that APDs could fully automate this process if they were to take over the prescription and administration of medicine as well, effectively removing humans from the loop. We propose that this should be a red line in the development and use of APDs—namely, decision-making and treatment should be left to humans, e.g., the farmer, farm advisor, or veterinarian. At the broader societal level, it needs to be discussed and determined which care tasks should be delegated to technologies, how much alienation between humans and animals we want to accept, and what limits should be placed on it.

We think the following set of principles is a valuable starting point to implement a morally acceptable and responsible usage of APD technologies (cf. [107]):

(1) The precautionary principle should be applied to err on the side of caution when animal well-being could be set back. This requires a nuanced and differentiated application of the principle. When identifying animal pain, it might be better to err on the side of caution—namely, to assume an animal is in pain when there is an indication it is, rather than assuming it does not feel pain. However, as overdiagnosing can come with severe negative consequences (see Sect. 3.1), it must be carefully decided what to conclude from it.
(2) Advances in animal pain detection due to APD must not be used as a reason to remove humans from the loop on decision-making in the context of animals' health and well-being.
(3) Individuals (e.g., farmers) should be held accountable for animal health or welfare problems that are detected by their APD systems but not investigated or treated.
(4) An open-source approach must be established for high-quality images to train APDs to make sure the technologies work as well as possible.

Overall, regulations are crucial for the responsible deployment of AI technologies, such as APDs, on farms. Therefore, we call for further research on the topic across disciplines and for the establishment of—ideally binding—regulations governing the use of AI to improve the well-being of farmed animals and to ensure these technologies serve animal interests rather than set them back.

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Declarations

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